

CHHAVI NAYYAR

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EDUCATION

University of British Columbia

Sept 2022 – May 2027

BSc, Combined Major in Computer Science, Biology, & Cognitive Systems

Vancouver, BC

- **Relevant Coursework:** Distributed Systems, Data Structures & Algorithms, Databases, Machine Learning, Software Engineering, Human-Centered AI, Cloud Infrastructure
- **Leadership:** Design Director – UBC UX Hub | TA – CPSC 532C (Human-Centered AI), COGS 300, COGS 303 | Member – Women in CS, Girls in STEAM

INDUSTRY EXPERIENCE

Software Engineer Intern

May 2026 – Aug 2026

Mastercard

Vancouver, BC

- Building microservices in **Java** and **Python** with **gRPC** and **RESTful APIs** for real-time transaction processing across payments infrastructure
- Deploying ML pipelines using **Docker**, **Terraform**, and **AWS** in a cloud-native production environment supporting fraud detection systems

Full-Stack Developer

July 2025 – Sept 2025

UBC Michael Smith Laboratories

Vancouver, BC

- Built responsive **TypeScript/React** site for the graduate Biochemistry program; established CI/CD workflows, cutting update turnaround by **30%** for 200+ stakeholders
- Conducted UX research and usability testing; implemented WCAG 2.1 accessibility improvements across navigation and information architecture

Software Engineer Intern

Jan 2024 – Mar 2024

TechyWeb Solutions

Remote

- Developed **Python** REST APIs and contributed to CI/CD pipelines in an Agile team; focused on performance testing and iterative backend improvement

RESEARCH EXPERIENCE

Teaching Assistant – Human-Centered AI & Cognitive Systems

Jan 2026 – Apr 2026

UBC Computer Science & Cognitive Systems Departments

Vancouver, BC

- Supported graduate instruction for CPSC 532C/554C (Human-Centered AI), facilitating seminars on algorithmic fairness, bias, and human-computer interaction
- Led lab sessions for COGS 300 (Designing Cognitive Systems) and COGS 303 (Research Methods), bridging cognitive science theory with applied AI implementation

Automation Engineer

Sept 2025 – Apr 2026

UBC Extended Learning

Vancouver, BC

- Built end-to-end pipeline in **Python** and **Google Apps Script** converting CSV survey data to structured PDF reports, eliminating **~70%** of manual processing
- Engineered automated notification and reminder workflows, reducing manual coordination overhead across course operations teams

Machine Learning Engineer

June 2025 – Dec 2025

BC Cancer

Vancouver, BC

- Engineered unsupervised ML models in **Python** and **scikit-learn** to analyze qualitative healthcare feedback, improving thematic pattern detection by **40%**
- Developed open-source concept mapping toolkit with **NetworkX** and **FastAPI**, cutting manual analysis time by **50%**; work contributed to co-first-authored publication
- Built interactive **Plotly** dashboards visualizing clustering results for clinical stakeholders and policymakers

Deep Learning Engineer

Apr 2025 – Dec 2025

UBC Department of Computer Science

Vancouver, BC

- Architected VTNet models in **TensorFlow** and **PyTorch** for eye-tracking analysis, achieving **25%** improvement in predictive accuracy through optimization
- Orchestrated experiments on **Slurm** HPC clusters, accelerating training pipelines by **30%** and enabling large-scale iteration

PUBLICATIONS

- **Nayyar, C.***, Xu, H.*, Hilbers, D., Avery, J., Raman, S., Bates, A.T., Conati, C., Fayaz-Bakhsh, A., & Nunez, J. (2026). *Towards clinical implementation of artificial intelligence in cancer care: Concept mapping analysis of provincial workshop findings*. *medRxiv*. doi:10.64898/2026.03.26.26349205 *co-first author

TECHNICAL PROJECTS

Your Search Box – Multi-Modal Search Service | *TypeScript, Next.js, Node.js, NLP* 2025 – Present

- Built a server-side rendered search platform supporting voice, image, and text input using **TypeScript** and **Next.js**; live commercial product
- Architected flexible adapter pattern for headless CMS integration (Shopify, WordPress) with environment-driven configuration for multi-tenant deployment

Healthcare Concept Mapping Toolkit | *Python, NetworkX, FastAPI, Plotly* 2025

- Open-source toolkit for qualitative healthcare research: MDS, hierarchical clustering, and thematic mapping from stakeholder input data
- Exposed via **FastAPI** endpoints with interactive **Plotly** visualizations; adopted by clinical research teams at BC Cancer

VR Recommendation System | *Python, TensorFlow, Flask* 2025

- Hybrid collaborative filtering and content-based engine using **TensorFlow** and cosine similarity; achieved 20–25% improvement in F1 score and precision
- **Flask** REST API with batch inference and query caching for high-frequency recommendation requests

CourseInsights – Survey Analysis Platform | *Python, Flask, Pandas* 2025

- Full-stack app automating course survey analysis end-to-end: CSV ingestion, **Pandas** transformation, and structured PDF report generation via **Flask**
- Deployed on Railway and actively used by UBC Extended Learning instructors; reduced reporting time from hours to minutes

Webability – Accessibility Widget | *JavaScript, CSS, WCAG/ADA* 2024

- WCAG/ADA-compliant widget with font scaling, high-contrast modes, keyboard navigation, and screen reader support; live at webability.io
- Optimized for minimal JS bundle size and zero layout shift; designed for drop-in integration across existing sites

TECHNICAL SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, C++, SQL, HTML/CSS
- **Backend & APIs:** Node.js, Express.js, Flask, FastAPI, gRPC, RESTful APIs, Microservices
- **Frontend:** React, Next.js, Angular, Tailwind CSS, Responsive Design
- **Data & Storage:** PostgreSQL, MySQL, MongoDB, NoSQL, Data Pipelines
- **Cloud & DevOps:** AWS (EC2, S3), Docker, Terraform, CI/CD (GitHub Actions), Linux, Slurm
- **ML/AI:** TensorFlow, PyTorch, scikit-learn, NetworkX, Plotly
- **Tools & Practices:** Git, Figma, Jira, Agile/Scrum, Test-Driven Development